

Chapter 16 — Fundamentals of Upper Urinary Tract Drainage

- Effective upper tract drainage is a **foundation of urologic practice**: it facilitates perioperative management of many upper tract conditions and is **essential in acute situations**.

INDICATIONS

Obstruction

- **The commonest indication mandating drainage**. Defined as blockage of urine flow through the intrarenal collecting system and ureter; **acute or chronic, partial or complete**.
- Presentation ranges from **silent, long-standing obstruction with loss of function** to **acute critical illness** needing urgent intervention.

Pattern	Causes
Acute	Stones, blood clots, sloughed papilla, fungal balls
Insidious	Tumours, ureteropelvic junction obstruction (UPJO), strictures
From the lower tract	Bladder tumour invading or occluding the orifices; outlet obstruction (benign prostatic hyperplasia, urethral stricture) — lower tract drainage usually corrects it , but upper tract drainage is sometimes required

Diagnostic Studies

- **Whitaker test**: percutaneous urodynamic study — irrigant infused at a **constant rate** into the kidney, **renal pelvic and bladder pressures** measured; a **differential above 12 cm water suggests obstruction**. Distinguishes obstructive from non-obstructive hydronephrosis.
- **Antegrade pyelogram** through renal access to evaluate ureteral strictures, often after **ureteral rest** (time for the stricture to mature after manipulation or catheter removal), which **improves surgical success**.

Perioperative Drainage

- **Preoperative percutaneous access** can expedite surgery for large renal stones or intrarenal tumours; **postoperative percutaneous drainage** for patients at higher risk of infection or needing **tamponade of bleeding**.
- **Preoperative ureteral stents** passively dilate the ureter before retrograde intrarenal surgery; **postoperative stents** mitigate ureteral oedema.

Mitigating Complications of Trauma

- Percutaneous or ureteral drainage **diverts urine and prevents urinoma**; ureteral or nephroureteral tubes limit the consequences of ureteral injury and stricture.

Instillation of Therapy

- Percutaneous access permits **instillation of chemotherapy for upper tract urothelial carcinoma** (newer agents may be chemoablative) and has been used for **percutaneous stone dissolution**.

Concept — drainage serves five purposes — **relieve, diagnose, prepare, protect, treat** — but obstruction dominates, and the choice between retrograde and antegrade is then set by anatomy.

ANATOMIC CONSIDERATIONS

- Feasibility and success depend on **patient-specific anatomy** (a pelvic kidney may be unsafe to puncture; diversions block retrograde access) and on history — abdominal surgery, prostatic enlargement, urethral stricture.

Horseshoe Kidney

- **Commonest renal fusion anomaly.** Fusion prevents medial rotation during ascent → **renal pelvis and ureter lie anteriorly**; the **high anterosuperior ureteral insertion** hampers retrograde access (stenting may still be attempted acutely). **UPJO coexists in one-third.**
- Percutaneous: **upper-pole calyces are easiest** — rarely supracostal, and reach the rest of the kidney and the ureter (Fig. 16.1); entry is **more medial**, often through the **paraspinous muscles**; the lower pole may be too distant → **extra-long instruments or mid-calyceal access.**
- **Axial imaging before access: retrorenal bowel is common.** Vessels are aberrant but enter **anteromedially** (except at the isthmus), so direct vascular injury is rare with planned access; pooled series show major haemorrhage uncommon and colon injury rare.

Ectopic Kidney

- Usually **pelvic**; also intra-abdominal or intrathoracic. Retrograde limited by UPJO or distance; posterior access blocked by pelvic bones → **supine, transabdominal** if no bowel intervenes.
- Options: **computed tomography (CT) guidance**; **laparoscopic or robotic mobilisation of bowel** (laparoscopy-assisted percutaneous nephrolithotomy (PCNL) in pelvic kidneys uniformly successful without complications in reported series); ultrasound to identify bowel; where no bone intervenes, posterior access **just above the iliac bone** or **transgluteal through the greater sciatic foramen.**

Transplant Kidney

- **Retrograde access is difficult: anterior ureteroneocystostomy** and **short ureter** give poor wire purchase → without retrograde opacification, **puncture under ultrasound.**
- **Pelvic location** → **anterior approach, supine**; **extraperitoneal**, so bowel rarely intervenes.
- **Dense perinephric fibrosis** resists balloon dilation → **fascial incising needle with balloon**, or **telescoping metal dilators.** Reported series: high technical success, no major complications.

Duplication

- **Complete** — separate ureters to the bladder; **incomplete** — ureters join below the ureteropelvic junction; **bifid** — moieties join at the renal pelvis. Both approaches possible; in complete duplication, **confirm the target moiety** for tube or stent.

Ureteropelvic Junction Obstruction

- Drain for pain, nausea, vomiting or **obstructive pyelonephritis.** **Antegrade preferred acutely when pyeloplasty is planned** — spares the ureter inflammation at repair; **retrograde stent after endopyelotomy** for healing.

Infundibular Stenosis/Diverticulum

- Stenosis (congenital or acquired) → **hydrocalycosis**; a diverticulum drains through a narrow channel. Acutely, **antegrade access**; retrograde requires **dilation or incision** of the neck.

Ureteral Surgery

Prior surgery	Access
Ureteral reimplantation	Retrograde, often with a flexible cystoscope

Prior surgery	Access
Ureterocalicostomy	Antegrade via upper pole — whole system reached, proximal ureter protected
Transureteroureterostomy	Antegrade — avoids retrograde manipulation
Ileal ureter	Antegrade — prevents stent migration

Urinary Diversion

- Retrograde is sometimes possible with a **flexible cystoscope** to find the ureteroenteric anastomoses, but **antegrade is more reliable, albeit more morbid**. Contrast injected into the diversion opacifies the kidneys (most anastomoses **reflux**); **non-refluxing or strictured → ultrasound-guided access**.

Body Habitus

- **Lower-limb contractures** hinder urethral access and positioning; **severe scoliosis** displaces perirenal organs.

KEY POINTS

- Access into a posterior calyx usually allows direct access to the remainder of the kidney.
- Upper-pole calyces are usually situated in a mediolateral orientation.
- Cross-sectional imaging is recommended before percutaneous access to evaluate perirenal anatomy.

Concept — anatomy chooses the route: **where the ureter is inaccessible go antegrade; where the posterior approach is blocked go anterior or image-guided**; cross-sectional imaging is the constant.

RETROGRADE RENAL ACCESS

Preoperative Considerations

- **Urinalysis for infection**. American Urological Association (AUA): **antibiotic prophylaxis for all cystoscopy with ureteral instrumentation**.
- **Critically ill patients** at risk of decompensation on induction: stent under **local anaesthesia, awake**; otherwise general anaesthesia or sedation.

Surgical Technique

- **Dorsal lithotomy** (flexible cystoscopes allow other positions) → ureteral catheter through the working channel into the orifice (a **guidewire** helps with distal stones, strictures, J-hooked ureters) → **retrograde pyelography** delineates ureter, collecting system and obstruction.

Ureteral Catheters and Guidewires

- **Catheters stabilise the working wire**; most are **straight, open-ended, 5–6 Fr**; angled-tip or torque-stabilised angiographic catheters for difficult cases.
- Guidewires: straight or curved tip; **polytetrafluoroethylene (PTFE)**, **hydrophilic-coated nitinol**, or **hybrid** (hydrophilic tip, PTFE body); stiffness from standard to **super-stiff**. First goal: **wire into the collecting system**.
- **Hydrophilic wires are most dependable for bypassing impacted stones**. Stiffer wires need more insertion force but prevent stent buckling in a **tortuous ureter or long dense stricture**: floppy wire and catheter navigated bend by bend into the kidney → exchange through a straight catheter for a stiff wire → ureter straightened → stent.

Stent Placement

- Stent passed **coaxially over the working wire** into the collecting system.

	Through the cystoscope (direct vision)	Fluoroscopy only
Sequence	Stent and pusher through working channel; distal end watched in bladder, proximal end on fluoroscopy → wire partly withdrawn, proximal curl forms → scope to bladder neck, pusher just in view → wire fully out, distal curl forms	Scope removed; pusher advances stent over wire (Fig. 16.2: through a 10-Fr sheath) → fluoroscope over symphysis, pusher's radio-opaque marker to mid-symphysis (male) or lower border of symphysis (female) → fluoroscope over kidney, wire partly withdrawn → proximal curl → back to symphysis, remove string keeping pusher still → wire out, distal curl
Advantages	No overpassing of the distal end; less fluoroscopy; cystoscope stability against buckling (stricture, impacted stone)	No scope needed during passage
Limits	—	Symphysis landmarks invalid after radical prostatectomy and with vaginal prolapse → spot image with scope at bladder neck, or place under direct vision

Stent Materials

- Most are **polymers (polyurethane, silicone)**, proprietary blends designed to resist **mineral deposition and biofilm**, i.e. encrustation.
- **Metallic stents resist extrinsic compression** (malignant obstruction): **Resonance** — tightly coiled metal alloy, first used in **2006 for retroperitoneal fibrosis**; durable patency and acceptable tolerability in **benign and malignant** obstruction.
- **Hybrid stents** (hydrophilic polymer reinforced with metal coils — Silhouette, Passage/Snake): high compression resistance in vitro, **limited clinical data**.

Stent Design

- The **ideal stent**: easy to insert, **memory against migration**, drains, **radio-opaque**, resists encrustation, minimal symptoms.
- **Self-retaining double-pigtail stents are most common** (Fig. 16.3). **Single-pigtail diversion catheters**: distal end exits through an ostomy. **Open-ended, spiral- or whistle-tip ureteral catheters**: temporary, to identify the ureter during abdominal or pelvic surgery.
- Novel distal ends to reduce bladder irritation (e.g. **Polaris Loop**): a randomised trial showed **no symptomatic benefit** over the double-pigtail on a validated questionnaire.

Stent Tolerance

- Symptoms arise from **intravesical stent material irritating the bladder** and **reflux of urine through the stent**. Consent should cover **urgency, frequency, dysuria, haematuria, urethral, groin and flank pain**.
- **Ureteral Stent Symptom Questionnaire (USSQ)**: validated; **six domains, 38 questions** — urinary symptoms, pain, general health, work performance, sexual function, additional problems. In its founding cohort most patients had urinary symptoms and activity-limiting pain, and a substantial minority sexual dysfunction and reduced work capacity.

Factor	Effect on stent symptoms
Diameter	No association
Length	Longer stents (22–26 cm) gave more urgency and frequency, same flank pain; a large randomised trial using the USSQ found no difference
Distal curl crossing the midline	Early link to urinary symptoms, not confirmed

Factor	Effect on stent symptoms
Intraureteral placement (only string crosses the orifice)	Improved symptoms — less intravesical material, better tolerance
Material	Hydrocoated silicone less pain and fewer urinary symptoms than non-silicone after ureteroscopy (multicentre randomised trial)
Drugs	α-blockers and antimuscarinics improve bladder symptoms (meta-analyses of small randomised trials)
Opioid-free enhanced recovery after surgery (ERAS)	Preoperative paracetamol, gabapentin, belladonna–opium suppository; intraoperative ketorolac; discharge on paracetamol, ibuprofen, oxybutynin and tamsulosin for at least 1 week to stent removal — pain and quality of life no worse

Complications

- **Misplacement:** proximal end left **below the obstruction**; stent pushed **submucosally** — suspect when wire, catheter or stent needs too much force → **repeat retrograde pyelogram** and renavigate. Parenchymal perforation and intravascular placement: very rare case reports.
- **Migration:** from anatomy or a **too-short stent**. **Obstructive symptoms with a stent in place** → **image** for malposition or failure.
- **Stent failure** — multifactorial: extrinsic compression, **infection with pseudocapsule**, debris, encrustation. Highest risk: **bilateral obstruction, severe hydronephrosis, recurrent UTI, intrinsic ureteral obstruction, prior radiation**. Long-term stenting may still be attempted with monitoring.
- **Encrustation:** urinary minerals deposit on the stent; risk rises with **dwell time, bacterial colonisation** and patient mineralisation factors → failure, **stent fracture, ureteral avulsion on removal**. **All long-term stents monitored** — imaging or office cystoscopy.

KEY POINTS

- Retrograde pyelography should be used to delineate the collecting system during stent placement.
- α-Blockers and antimuscarinics can improve ureteral stent tolerability.
- Obstructive symptoms with an indwelling ureteral stent warrant evaluation for stent failure.
- Long-term indwelling stents should be monitored for encrustation.

Concept — *stent symptoms come mainly from **material in the bladder and reflux**, so what helps is less intravesical stent, softer material and bladder-directed drugs — **not a thinner stent**.*

ANTEGRADE RENAL ACCESS

Preoperative Considerations

- **Imaging.** Chosen by pathology: **cross-sectional imaging** for intra- and perirenal anatomy; **ultrasound suffices for simple decompression** of presumed distal obstruction; AUA stone guideline: **non-contrast CT before PCNL**; pyelography for calyceal anatomy; **nuclear renogram** when ipsilateral function is uncertain.
- **Antimicrobial prophylaxis.** Upper tract drainage is **clean-contaminated**; avoid antibiotic overexposure (resistance). Before any procedure: dipstick, with possible urinalysis and conditional culture.
 - **Culture required** with risk factors: prior UTI, **suspected struvite stone**, anatomic abnormality (UPJO, stricture), **indwelling tubes or stents**, urinary reconstruction (diversion, augmentation). A culture from an externalised drain is sometimes necessary but contamination-prone. Indwelling foreign bodies are colonised → **higher postoperative infection**.

- **AUA: periprocedural prophylaxis for ALL percutaneous renal surgery**, even with negative culture and low risk; a **short course (≤ 24 hours)** at nephrostomy or stent removal may be considered. Need for simple percutaneous drainage: **uncertain**.
- Cover **urinary and skin organisms**: first- or second-generation cephalosporin; aminoglycoside (**aztreonam in renal insufficiency**) plus metronidazole or clindamycin; ampicillin/sulbactam; fluoroquinolone.

Anticoagulation and Antiplatelet Management

- Careful review is **required before percutaneous access** (less critical for retrograde stenting); agree strategy with the prescriber; **bridging** is sometimes needed.

Agent	Stop before percutaneous access
Aspirin	1 week — for primary prevention only; continue in established heart, cerebrovascular disease or high cardiovascular risk
Warfarin	1 week
Clopidogrel	5 days
Non-steroidal anti-inflammatory drugs (NSAIDs)	3–5 days
Novel agents	Varying pharmacokinetics — involve a clinical pharmacist (International Consultation on Urological Diseases (ICUD)/AUA)

- **Uninterrupted aspirin during PCNL did not increase blood loss or transfusion**, including in high cardiovascular-risk patients (large retrospective series).

Anesthetic Considerations

- Set by the anaesthetic team from cardiopulmonary comorbidity and body mass index (BMI). **Prone** → **general endotracheal anaesthesia** (secure airway; **controlled ventilation** helps with a mobile kidney); **supine** → **laryngeal mask airway** possible; spinal anaesthesia is safe in some cases.
- Postoperative: **tract infiltration with local anaesthetic reduces analgesic need and 24-hour pain** (meta-analysis of randomised trials); **intercostal and thoracic paravertebral blocks** give similar benefit.

Choice of Calyx for Access

- **Treatment goal decides**: septic obstruction → **any easily accessed calyx** for swift decompression; preoperative drainage for a **staghorn** → a calyx serving the whole system for later PCNL. With interventional radiology, agree the approach in advance.
- Anterior or posterior calyces can be approached from different positions. **Never access directly through the renal pelvis**; beware **atrophic parenchyma** — the tract may not seal.

Calyx	Reach	Cost and cautions
Upper pole	Renal pelvis, proximal ureter, lower pole, often interpolar — rigid and flexible	Often supracostal → pleural risk . Above the 12th rib (11th intercostal space): benefit usually exceeds risk . Above the 11th rib: greater pleural and lung risk → displace kidney inferiorly: cephalad tilt of a subcostal sheath or needle in a lower calyx; traction on a through-and-through wire via lower pole; puncture at full inspiration
Interpolar	Renal pelvis, upper ureter	Least versatile — rigid instruments cannot reach the poles; reserve for interpolar work

Calyx	Reach	Cost and cautions
Lower pole	Most of the kidney; rigid to pelvis easily; flexible to most calyces	No pulmonary morbidity ; iliac crest may limit upward deflection

- Fig. 16.4: the supracostal approach gives **more direct access and a better angle** to the rest of the kidney than subcostal access to the same calyx.

Patient Positioning

	Prone	Supine
Origin	1955 (Goodwin); most used	1987 (Valdivia Uria); variants: Galdakao-modified supine, complete supine
Set-up	Chest on one horizontal roll across the nipple line or two vertical rolls along the mid-clavicular lines; head on pillows, neck neutral; arms in modified swimmer's position on armboards (brachial plexus); pad every pressure point; all radiolucent (Fig. 16.5)	Ipsilateral side at the table edge; flank raised on a bolster or 3-litre saline bag under the lumbar fossa; ipsilateral arm across the chest; elbow and wrist padded
Advantages	Broad surface ; any pole; multiple tracts or bilateral without repositioning; upper pole closest to skin ; with free abdominal excursion, higher pulmonary capacity, same peak inspiratory pressure	Airway easy ; better cardiopulmonary function; no repositioning → faster PCNL (randomised trials); less hand radiation ; surgeon seated
Disadvantages	Airway harder; lower cardiac index ; impossible with spinal deformity or super-morbid obesity; pressure injury (rhabdomyolysis, neuraxial injury); ocular injury unique to prone ; surgeon fatigue	Limited puncture surface ; upper pole difficult ; low intrarenal pressure (sheath points down, drains) → poorer vision

Modifications to Prone or Supine Positioning

- **Prone split-leg or supine with modified lithotomy** give urethral access for **combined antegrade-retrograde** surgery.

Surgical Technique

Image Guidance for Puncture

Guidance	Strengths	Limits and settings
Fluoroscopy	Most common; delineates collecting system and pathology; wide field (collimate to cut dose); monitors every step	Monoplanar → rotate the C-arm to reveal 3D (three-dimensional) configuration; retrograde air and/or contrast enhances it — so depends on a retrograde catheter
Ultrasound	No ionising radiation , cheaper; shows adjacent viscera; Doppler shows vessels; clearly separates anterior from posterior calyces ; proven in children, obesity, aberrant anatomy	Needs no retrograde catheter → ideal after ileal conduit or orthotopic diversion . Curved array , typically 3.5 MHz; depth 8–12 cm; raise gain for stone-needle-system contrast; etched needle keeps tip visible (Fig. 16.6)
CT or MRI	Images bowel and viscera; technique as for percutaneous biopsy	Complex or dangerous anatomy — pelvic or ectopic kidneys
Electromagnetic, robotic	Described	Clinical role still being defined

Concept — every antegrade decision balances **reach against morbidity**: the upper pole reaches most but risks the pleura, prone exposes most but burdens the airway, and fluoroscopy shows the system while ultrasound shows what surrounds it.

EQUIPMENT

Access Needles

- **Seldinger technique** (1953, for vascular access): needle through skin into the collecting system → guidewire through needle → devices over the wire → tract enlarged gradually to the required lumen.

	21-gauge	18-gauge
Wire	0.018-inch — must be exchanged for 0.035-inch	Standard 0.035-inch directly
Trauma	Less; multiple passes with minimal bleeding risk	More; avoid multiple passes
Trajectory	Deflects in scarred kidneys and obesity	Stiffer — holds trajectory
Drawback	Exchange adds a step and risk of losing access	—

- Both have a **blunt sheath and sharp obturator** (Fig. 16.7); the clinical superiority of either is uncertain.
- **Upsizing 0.018 → 0.035 inch: coaxial introducer** — inner catheter and 0.018 wire removed, 0.035 wire passed through the outer catheter; or **graduated introducer** — tip fits the 0.018 wire, lumen widens, and a **side hole a few centimetres back** admits the 0.035 wire. A **stiffener** helps either pass over the flimsy 0.018 wire.

Guidewires

- Goal: **secure position** for instruments and tubes — ideally **down the ureter into the bladder** to prevent displacement.
- **Safest initial wire: PTFE-coated J- or C-wire** — curved tip rarely perforates, but **will not easily pass down the ureter** → floppy-tip PTFE or hydrophilic (straight or angled) wire for that.
- **Upper-pole puncture is in line with the ureter** — wire often passes unaided; from a lower pole, or on failure, a **torque-stabilised angled catheter** steers it.
- Then exchange for a **stiff wire (Amplatz Super Stiff)** for dilation, and place a **second safety wire** via a dual-lumen or larger coaxial catheter.
- **Single wire acceptable** when a second is impossible — complete staghorn, calyceal diverticulum, impacted infundibular stone — but keep trying to secure the ureter.

Tract Dilation

- Incise the skin to the **full final sheath size** first. Cardinal principle: **do not pass the dilator too far** — perforation of the infundibulum, renal pelvis or ureteropelvic junction; **dilate short rather than long** — a short sheath can be advanced later.

DILATORS COMPARED

	Metal (Alken)	Plastic (Amplatz)	Balloon	Single-step
Design	Coaxial stainless-steel rods over an 8-Fr guide rod with a ball tip set at intended depth (Fig. 16.8)	Progressively larger polyurethane dilators over an 8-Fr PTFE guiding catheter ; each removed before the next (Fig. 16.9)	Sheath back-loaded; radio-opaque marker at depth ; pressure-syringe inflation (Figs. 16.10–16.11)	Final plastic dilator alone; balloon or rigid dilator with expandable sheath; small dilators in mini-PCNL
Strength	Most effective; dilates dense scar ; reusable, cheap	Thought less traumatic	Most used today ; single pass; better with hypermobile kidney ; less haemorrhage and transfusion (most series, meta-analysis)	Shorter access and fluoroscopy times without more complications (meta-analysis of randomised trials), any position

	Metal (Alken)	Plastic (Amplatz)	Balloon	Single-step
Weakness	Can cause considerable damage; depth hard to hold against scar (tapered, centimetre-marked modification)	Bleeding at each withdrawal; disposable, dearer; complications no different from metal	Expensive, single-use; less effective in dense scar (PTFE-reinforced high-pressure balloons may help)	—

- Rigid dilators must be advanced **in line with the access needle** — they kink and displace the wire — under fluoroscopy.
- Balloon: a **"waist"** appears at resistance, usually **abdominal wall fascia and renal capsule**; the **"shoulder"** lies between balloon end and point of maximal diameter — **advancing the sheath beyond maximal diameter risks collecting system injury**.
- The only randomised trial of all four methods: **no difference** in access time, haemoglobin fall, complications or success; **fluoroscopy time lower with balloon and single-step**.
- **Commonest cause of difficult dilation: prior renal surgery and perirenal scarring**. If rigid dilators fail: **Collings knife, atherotomy cutting balloon, fascial incising needle**. Repeat PCNL in scarred kidneys usually succeeds.

Concept — dilator choice is a **scar-versus-mobility trade**: rigid metal wins in scar, balloon wins in a mobile kidney and bleeds less; the rule common to all is **never beyond the target calyx**.

POSTPROCEDURAL DRAINAGE

General Considerations

- Nephrostomy tubes give good drainage and control while preserving access; particularly useful with obstruction plus bowel diversion, impacted stones precluding a stent, or extrinsic compression causing stent failure.
- Tube size (prospective and randomised trials, 20–24 Fr vs 8–18 Fr): **smaller tubes → less pain and less leakage, no more bleeding**. Larger tubes occasionally bleed **immediately on removal** → if bleeding is feared, **remove over a wire or in the radiology suite** (replacement, embolisation); most are removed safely at the bedside. **Clamp before removal** after percutaneous surgery to confirm distal patency.
- **Combined nephrostomy + ureteral stent**: ureteral injury or oedema (drainage, less stricture); after **complex reconstruction** (leak risk; upper tract control after stent removal); significant urinary tract trauma.

Management of Nephrostomy Tube

- Secure with **external sutures or dedicated adhesive dressings** against dislodgement. Externalisation → colonisation and infection → **routine exchange every 3 months**; encrustation can block tubes. **Flushing and hydration** maintain patency but risk introducing infection without strict sterile technique.

Types of Catheters

Catheter	Retention	Size	Features and cautions
Balloon (Foley, Councill; Fig. 16.12)	Balloon — partly filled (full balloon too large for most systems)	Up to 24 Fr	Inflate with saline or water, not contrast (slows deflation, spoils imaging); Councill end hole admits a small catheter down the ureter
Cope (Fig. 16.13)	String-locked loop coiled in the system	6–14 Fr	Small bore (8–10 Fr) after a large PCNL tract may dislodge; main dislodgement risk factor: high BMI ; place loop in a polar

Catheter	Retention	Size	Features and cautions
			calyx rather than pelvis ; with a pigtail stent, keep loop away from stent coil and remove over a wire (no interlocking)
Nephroureteral stent (Fig. 16.14)	Cope-type renal coil + passive bladder pigtail	8–12 Fr; 20–28 cm	Excellent control of the whole upper tract , easy re-access, rarely dislodged
Malecot (Fig. 16.15)	Expanded wings — modest, atraumatic	16–30 Fr	Stiffener needed for placement and removal (straighten tube so it aligns); re-entry catheter : 18-cm ureteral extension — wings can be externalised while a wire is passed down the ureter
Circle	Loop through two tracts	—	Secure, easily exchanged, atraumatic, excellent drainage, renal pelvic irrigation ; radio-opaque markers keep drainage holes inside; needs a Y-connector

- **All nephrostomy tubes are secured externally.** The best argument for a tube that **extends down the ureter** is that it keeps a conduit into the collecting system **even if the renal portion is pulled out**, and allows **rapid replacement to tamponade bleeding**.
- **Nephroureteral stent placement:** advanced over a wire ending in the bladder → renal coil side holes positioned fluoroscopically → bladder end coiled generously, wire removed → catheter moved in and out while **pulling the string and rotating the external portion clockwise** → renal coil forms in the pelvis.
- **Circle catheter placement:** access two distinct calyces, wires through both → flexible endoscope through one tract grasps the wire from the other and externalises it → catheter guided along it. Second-look re-access: wire through the catheter re-establishes through-and-through access (if resistance, advance catheter and wire together).

"Tubeless" With Ureteral Stent

- A nephrostomy is **not mandatory**. Omitting it **reduces pain, analgesic use, hospital stay and cost**; a short-term ureteral stent (± string) is usually left.
- Evidence (meta-analyses of randomised trials): **shorter stay and operative time** tubeless; benefit in stay and pain is clear against **large tubes (>16 Fr)** but **not against 8–9 Fr tubes**; **no difference in fever/infection, transfusion or stone-free rates**. Safe with **supracostal access** and **bilateral simultaneous** procedures.
- Disadvantages: **loss of the tract** for a second look; **stent discomfort** — worse Wisconsin Stone Quality of Life (WiSQoL) scores than nephrostomy in two randomised trials.
- **Externalised stents** (single pigtail through the urethra, or string out through the flank) are removed before discharge without a procedure; safe; one trial vs large-bore tube: **less narcotic, shorter stay**; internal vs external stents: equal except **more stent symptoms with internal**.
- **Keep a nephrostomy** when external drainage is needed: **pyonephrosis, significant bleeding, significant collecting system injury**.

Concept — the "tubeless" advantage is really an advantage over **large tubes**; against a small nephrostomy it largely disappears, and it trades the tube's discomfort for the stent's.

Complications

- Percutaneous renal surgery can be morbid and rarely fatal: **slightly over 10% have major complications** (modified Clavien–Dindo).

Acute Hemorrhage

- The kidney's rich blood supply makes haemorrhage **the most significant complication of percutaneous access**: nephrostomy alone needs **transfusion in 0.5–4%**.

- Most intraoperative bleeding is **parenchymal and insignificant**. Management ladder: venous → **pressure on the incision** → persistent tract bleeding in a stable patient → **occlude the nephrostomy** so the system clots and tamponades → gentle irrigation next day → continued → **imaging/angiography**. Intraoperative arterial bleeding: **electrocautery** sometimes; otherwise **angioembolisation**.

Delayed Hemorrhage

- Unrecognised intraparenchymal vascular injury → **pseudoaneurysm or arteriovenous fistula (AVF)**; treatment needed in **under 1%** of major percutaneous procedures; bleeding usually at **about 1 week**.
- Mechanisms, as a pair: **pseudoaneurysm — a clot temporises the arterial injury and ruptures intermittently**; **AVF — adjacent artery and vein heal together, and the thin vein cannot withstand arterial pressure**.
- **Any bright-red blood in urine or tube after surgery** raises suspicion; flank pain from clot obstruction or expanding haematoma. **Angiography is diagnostic in over 90%** and allows **selective embolisation** in the same sitting; may miss bleeds with multiple tracts or intermittent bleeding; **non-selective embolisation risks partial or total renal loss**.
- Figures: Fig. 16.16 — renal artery injection filling the **inferior vena cava early = AVF**; Fig. 16.17 — lower-pole pseudoaneurysm occluded, but embolisation **devascularised much of the lower pole**.

Systemic Inflammatory Response Syndrome and Sepsis

- **Sepsis in 0.5–2.5% after PCNL**; much less after uncomplicated drainage alone.

Collecting System Injury

- **Infundibular tears**: usually insignificant, may bleed. **Renal pelvic injury** at access, dilation or manipulation: recognised by **contrast extravasation**, loss of pelvic distension, or direct view (Fig. 16.18) → **nephrostogram before tube removal** to confirm healing.
- Ureteral injuries mostly **heal with a stent**. Any injury can leak → infection, ileus; **symptomatic urinoma** → **percutaneous drain**; intra-abdominal extravasation reported.

Visceral Injury

- **Colon injury about 0.1% in PCNL**. **Left twice as often as right** — colonic apposition greatest at the **left lower pole**. Risk factors: **older age, dilated colon, prior colonic surgery or disease, thin habitus, horseshoe kidney**. Retrorenal colon on CT about 2% supine vs 10% prone → **supine may be safer**.
- Management: recognised intraoperatively → **separate drainage of colon and collecting system** (urinary drainage alone reported effective); **extraperitoneal** → **conservative** (nil by mouth several days; watch for infection or fistula when diet resumes); **missed injury** — unexplained fever, leucocytosis, prolonged ileus, peritonism; **intraperitoneal, peritonitis or sepsis** → **open repair**.
- **Small bowel (duodenum)**: uncommon; peritonitis or **nephroenteric fistula**; often open repair, though conservative management (percutaneous intraduodenal catheter or nasogastric/nasoduodenal drainage, upper tract drainage, fasting, parenteral nutrition) has succeeded.
- **Spleen and liver**: unlikely unless **supracostal access** (mostly above the 10th rib; possible above 11th/12th) or **organomegaly** → CT-guided access when imaging shows risk. Conservative: serial examination and full blood count; rarely laparotomy and **splenectomy**; liver bleeds less (deliberate transhepatic PCNL reported); tube removal in the interventional suite if bleeding is feared.
- **Gallbladder/biliary injury**: rare but needs **laparotomy/laparoscopy and cholecystectomy** — biliary peritonitis carries high mortality. Postoperative **pancreatitis without direct injury** described.

Pleural Injury

- Hydrothorax or pneumothorax: **supracostal access is the main risk**; below the 12th rib rare. Needing intervention: **above 12th rib about 5% (range 0–18%)**; **above 11th rib about 25% (range 0–100%)**.

- **Nephropleural fistula:** never with subcostal access; about 2% above the 12th and 6% above the 11th; persists because of **distal ureteral obstruction**.
- Intraoperative detection on fluoroscopy: **fluid tracking along the lateral chest wall, compressing the lung** → **small-calibre (8–12 Fr) Cope tube as thoracostomy** (Fig. 16.19), placed with the same fluoroscopic technique, on a **Heimlich valve** — water seal unnecessary without lung injury.
- Postoperative: **observe if no respiratory compromise or instability**; otherwise small-calibre tube; large-bore only for lung injury. Chest radiography is more sensitive, but a **clear intraoperative chest fluoroscopy** reportedly predicts no need for thoracostomy.

Neuromusculoskeletal Complications

- Pressure on nerves and vessels, from the table or limb position, causes short- or long-term disability: **support all pressure points; straps must not constrict**.
- **Prone carries the highest risk**, mostly head and neck: **ocular injury with visual loss**, facial nerve injury, necrosis over facial bones or nose tip; **carotid or vertebralbasilar dissection** → **stroke** → head **neutral, not extended**.
- Limbs: **shoulder and elbow abduction $\leq 90^\circ$** (brachial plexopathy); pad elbow, forearm and knees; **elevate ankles** off the dorsum of the foot.

Venous Thromboembolism

- **Low incidence ($\leq 3\%$)**; the AUA best-practice statement does **not list percutaneous renal surgery** among procedures needing thromboprophylaxis; **early ambulation** is the best measure.
- **Venous gas embolism:** rare, potentially fatal. Air enters veins → right heart → pulmonary circulation, **blocking right ventricular output** → hypoxaemia, hypercapnia, low cardiac output; through a **patent foramen ovale** → arterial embolism, neurological deficit. Associated with **air pyelography**, with or before puncture.
- Signs: hypoxaemia, pulmonary oedema, raised airway pressure, hypotension, jugular venous distension, facial plethora, dysrhythmia, **mill-wheel murmur**, wide QRS with right heart strain. **Most sensitive: sudden fall in end-tidal CO_2** .
- Treatment: **rapid ventilation with 100% oxygen, head down, right side up**, general resuscitation.

Concept — *bleeding is the signature complication and **timing sorts the source**: immediate bleeding is parenchymal or venous and settles with tamponade; bleeding at about a week is arterial (pseudoaneurysm or AVF) and needs angiography.*

KEY POINTS

- Antibiotic prophylaxis is recommended for all patients undergoing percutaneous renal surgery.
- Safe fluoroscopic access requires imaging in two planes to determine the 3D configuration of the collecting system.
- Ultrasound-guided access allows real-time identification of perirenal anatomy.
- "Tubeless" procedures may improve hospital stay and postoperative pain.
- Percutaneous access can be morbid; surgeons should understand the commonest complications and their management.